G H Raisoni International Skill Tech University, Pune.

Question Paper

Roll No.:

Continuous Assessment Examination (CAE)

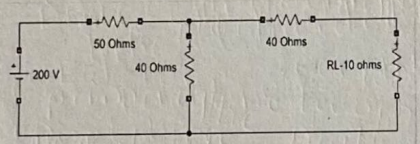
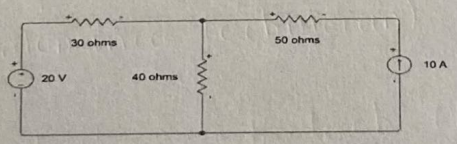
Programme: B Tech **Year: First** **Semester: I**
Course Code: MT103 **Max Marks: 20**
Course Name: Basic Electrical Engineering **Duration: 60 mins.**

Instructions:

1. You are permitted to use **Scientific Calculators** during the exam.
2. Utilize diagrams and equations where relevant, and always include units
3. Students must label their answers with the corresponding question number,such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	Demonstrate knowledge of fundamental electrical concepts such as voltage, current, resistance, power, and energy.
CO2	Apply circuit analysis techniques to analyze and solve simple DC and AC circuits, including series and parallel configurations.
CO3	Solve problems related to voltage, current, resistance, and power in electrical circuits using appropriate mathematical methods and principles.

Q. No.	Sub Q.No.	Question	Marks
1	Solve any **TWO**		
	A	Find Load current through 10 ohm resistor using Thevenin's Theorem. 	05
	B	I . Define Norton's Theorem and also explain it's steps to solve Norton's. II. State and explain KVL and KCL using an Example.	05
	C	Find current through 40 ohm resistor using S.P.T(SuperpositionTheorem) 	05
	D	I. Define Superposition theorem also Write it's steps to solvesuperposition theorem. II. Explain different types of Electrical Sources with flow chart.	05
2	Solve any **TWO**		
	A	Explain Brief about real, reactive and apparent power. Also given an example.	05
	B	State and Explain the representation of AC Sinusoidal Waveform.	05
	C	Derive the RMS (Root Mean Square) and Average values for voltage and current with derivation.	05
	D	Explain the mathematical derivation for the energy stored in a capacitorand inductor also write it's statement.	05